UNIVERSITI TUNKU ABDUL RAHMAN

UCCD 2213 SOFTWARE ENGINEERING PRINCIPLES

FEB TRIMESTER 2025

ASSIGNMENT

A. TOPIC COVERED

This assignment emphasizes more on architectural pattern design. Students shall apply their lesson learned for the aforementioned topic in the lecture and tutorial. For further information, you may refer to slides.

B. ASSIGNMENT INSTRUCTIONS

1) ASSIGNMENT MARKS

The total of this assignment is **100 marks**, which only contributes **15 marks** of your total coursework marks.

2) GROUP MEMBERS

It’s a **Group Assignment**. Each group can only have **FOUR (4) students**. It is **NOT ALLOWED** to form a group with more than FOUR (4) members. All members in a group shall participate and complete the **Individual and Group Parts** in this assignment.

3) FULL REPORT FORMATS

Students shall be responsible to ensure that all the following formats are fulfilled before uploading the full report into the Google Form link given into the WBLE portal AND a hardcopy to the lecturer.

Font Type	Times New Roman
Font Size	12
Line Spacing	1.5
Header	UCCD2213 Software Engineering Principles (10 font size)
Footer	Page Number (on Right Corner, 10 font size)
Save As	.pdf
File Name	Assignment Title

4) **ASSIGNMENT FRONT COVER**

It's compulsory to use the **FRONT COVER** as shown in appendix 1 in this document. All members shall provide **DIGITAL SIGNATURE** on the front cover.

5) **ASSIGNMENT DEADLINE**

The assignment full report is due on **Week 10 (11 April 2025) by 6pm**. The **SOFTCOPY** of full report for this assignment must be uploaded in the Google Form link that will be given in the WBLE during week 10 and a **HARDCOPY** to the lecturer at her office [NG-038] between 11am-3pm.

6) **PENALTY OF LATE SUBMISSION**

Late submission will deduct 15% of total assignment marks for assignment report.

7) **PLAGIARISM**

The evidence of plagiarism or collusion will be taken seriously, and University regulations will be applied fully to such case. You are advised to be familiar with the University's definitions of plagiarism and collusion. Any plagiarism or collusion may result in disciplinary action, in addition to **ZERO MARK** will be awarded to all parties involved.

C. GROUP & INDIVIDUAL TASKS

1) **REQUIREMENTS**

In this assignment, you are required to propose an **Assignment Title**, which must relate to **Business Information System (BIS)**. The system shall be able to make business growth and help the staff to be more productive to deal with their daily routine. The objective is to analyze a given problem, propose an appropriate architectural solution, and justify the application of design patterns in addressing system requirements and challenges.

2) QUESTIONS

For this Part I, you need to work INDIVIDUALLY.

PART I: Design a system that caters to the following requirements: **(20 marks)**

- User Profile Management: Secure user authentication and account management.
- Data Integration: Collect and aggregate data from multiple external and internal sources.
- Personalized Recommendations: Provide tailored suggestions based on user data and behavior patterns.
- Data Privacy and Security: Ensure strict compliance with data protection regulations and best practices.
- Scalability Requirements: Support a growing user base and evolving system features.

You are tasked to design an architectural solution for this system using pattern-based techniques. The design must accommodate flexibility, scalability, and maintainability while addressing the specified requirements.

PART II: Answer the following questions from Q1-Q4.

For this Part II, you need to work in group of four.

Q1: Requirement Analysis and Design Principles **(20 marks)**

- Analyze the key functional and non-functional requirements for your chosen system.
- How would design principles such as modularity, scalability, maintainability, and security guide your architectural solution?

Q2: Pattern Selection and Justification **(20 marks)**

- Select at least three appropriate design patterns to address the system's requirements.
- Provide a detailed justification for the selection of each pattern.
- Explain how these patterns contribute to solving specific design challenges within the system.

Q3: Component and Architecture Design **(25 marks)**

- Develop a high-level architectural design for your system.
- Include the following elements:
  - Component Diagram: Depict the main components and their interactions.
  - Component Responsibilities: Explain the role and functionality of each component.
  - Pattern Integration: Show how the chosen patterns are incorporated into the architecture.

For this Part II (Q4), you need to work INDIVIDUALLY.

Q4: Reflection on Design Choices **(15 marks)**

- Reflect on how applying pattern-based techniques influenced your architectural decisions.
- What lessons did you learn from using design patterns in this assignment?

D. ASSIGNMENT SCHEDULES

Week 2	: Form groups and propose assignment title
Week 3 & 4	: Part 1 – Project Introduction and Architectural Designing
Week 5 & 6	: Part 1 – Project Introduction and Architectural Designing
Week 7 & 8	: Part 2 – Project explanation and in-detail explanation of the project.
Week 9 & 10	: Assignment Fine Tune & Assignment submission

Appendix 1: Cover page

UCCD2213 SOFTWARE ENGINEERING PRINCIPLES

#Assignment Title

No	Student Name	Student ID	Total Mark
1			
2			
3			
4			

No	Assessment Components	Assessment Marks		Students' Marks				
		Group	Individual	Group	Individual			
					1	2	3	4
1.0	**PART 1 (20 marks)**							
1.1	Design an architectural design for the system.	-	20	-				
2.0	**PART 2 (80 marks)**							
2.1	Q1: Requirement Analysis and Design Principles	20	-		-	-	-	-
2.2	Q2: Pattern Selection and Justification	20	-		-	-	-	-
2.3	Q3: Component and Architecture Design	25	-		-	-	-	-
2.4	Q4: Reflection on Design Choices		15	-				
	TOTAL	100						